

Qunatim Analytics

Task -2

This code is written in python.

Importing library and dataset

```
import pandas as pd
```

```
import numpy as np
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
import warnings
```

```
warnings.filterwarnings('ignore')
```

```
df = pd.read_csv("QVI_data.csv")
```

using datetime function

```
df["DATE"] = pd.to_datetime(df["DATE"])
```

extracting trail store data

```
trail_store = df[(df["STORE_NBR"]==77)|(df["STORE_NBR"]==86)|(df["STORE_NBR"]==88)]
```

```
trail_time_period = trail_store[(trail_store["DATE"]>="2019-02-01")&(trail_store["DATE"]<="2019-04-30")]
```

#generting new column called month

```
trail_time_period['month'] = trail_time_period["DATE"].dt.strftime("%b")
```

how much sell in every month

```
every_month_sales = pd.DataFrame({
```

```
    "month":['Feb','Mar','Apr'],
```

```
"store_77":[trail_time_period[(trail_time_period["STORE_NBR"]==77)&(trail_time_period["month"]=='Feb')][TOT_SALES'].sum(),
```

```
trail_time_period[(trail_time_period["STORE_NBR"]==77)&(trail_time_period["month"]=="Mar")][TOT_SALES'].sum(),
```

```
trail_time_period[(trail_time_period["STORE_NBR"]==77)&(trail_time_period["month"]=="Apr")][TOT_SALES'].sum()],
```

```
"store_86":[trail_time_period[(trail_time_period["STORE_NBR"]==86)&(trail_time_period["month"]=='Feb')][TOT_SALES'].sum(),
```

```
trail_time_period[(trail_time_period["STORE_NBR"]==86)&(trail_time_period["month"]=="Mar")][TOT_SALES'].sum(),
```

```
trail_time_period[(trail_time_period["STORE_NBR"]==86)&(trail_time_period["month"]=="Apr")][TOT_SALES'].sum()],
```

```
"store_88":[trail_time_period[(trail_time_period["STORE_NBR"]==88)&(trail_time_period["month"]=='Feb')][TOT_SALES'].sum(),
```

```
trail_time_period[(trail_time_period["STORE_NBR"]==88)&(trail_time_period["month"]=="Mar")][TOT_SALES'].sum(),
```

```
trail_time_period[(trail_time_period["STORE_NBR"]==88)&(trail_time_period["month"]=="Apr")][TOT_SALES'].sum()])
```

```
})
```

```
Print(every_month_sales)
```

```
# Then plotting sales trend in every month
```

```
#store 77
```

```
fig,ax = plt.subplots(3,1,figsize=(12,9))
```

```
fig.suptitle("Store 77",fontsize=16)
```

```
ax[0].plot(trail_time_period[(trail_time_period["STORE_NBR"]==77)&(trail_time_period["month"]=='Feb')].sort_values(by='DATE')['DATE'],trail_time_period[(trail_time_period["STORE_NBR"]==7
```

```
7)&(trail_time_period["month"]=="Feb')).sort_values(by='DATE')['TOT_SALES'], marker='o',  
linestyle='-',color='purple',label='Feb')
```

```
ax[0].grid(True)
```

```
ax[0].legend(loc='upper left')
```

```
ax[1].plot(trail_time_period[(trail_time_period["STORE_NBR"]==77)&(trail_time_period["month"]  
=='Mar')].sort_values(by='DATE')['DATE'],trail_time_period[(trail_time_period["STORE_NBR"]==7  
7)&(trail_time_period["month"]=="Mar")].sort_values(by='DATE')['TOT_SALES'], marker='o',  
linestyle='-',color="orange",label="Mar")
```

```
ax[1].grid(True)
```

```
ax[1].legend(loc='upper left')
```

```
ax[2].plot(trail_time_period[(trail_time_period["STORE_NBR"]==77)&(trail_time_period["month"]  
=='Apr')].sort_values(by='DATE')['DATE'],trail_time_period[(trail_time_period["STORE_NBR"]==77  
)&(trail_time_period["month"]=="Apr")].sort_values(by='DATE')['TOT_SALES'], marker='o',  
linestyle='-',label="Apr")
```

```
ax[2].grid(True)
```

```
ax[2].legend(loc='upper left')
```

```
plt.tight_layout()
```

```
plt.show()
```



```
#store 86
```

```
fig,ax = plt.subplots(3,1,figsize=(12,9))
```

```
fig.suptitle("Store 86",fontsize=16)
```

```
ax[0].plot(trail_time_period[(trail_time_period["STORE_NBR"]==86)&(trail_time_period["month"]  
=='Feb')].sort_values(by='DATE')['DATE'],trail_time_period[(trail_time_period["STORE_NBR"]==8  
6)&(trail_time_period["month"]=="Feb")].sort_values(by='DATE')['TOT_SALES'], marker='o',  
linestyle='-',color='purple',label='Feb')
```

```
ax[0].grid(True)
```

```
ax[0].legend(loc='upper left')
```

```
ax[1].plot(trail_time_period[(trail_time_period["STORE_NBR"]==86)&(trail_time_period["month"]  
=='Mar')].sort_values(by='DATE')['DATE'],trail_time_period[(trail_time_period["STORE_NBR"]==8
```

```
6)&(trail_time_period["month"]=='Mar')).sort_values(by='DATE')['TOT_SALES'], marker='o',  
linestyle='-',color="orange",label="Mar")
```

```
ax[1].grid(True)
```

```
ax[1].legend(loc='upper left')
```

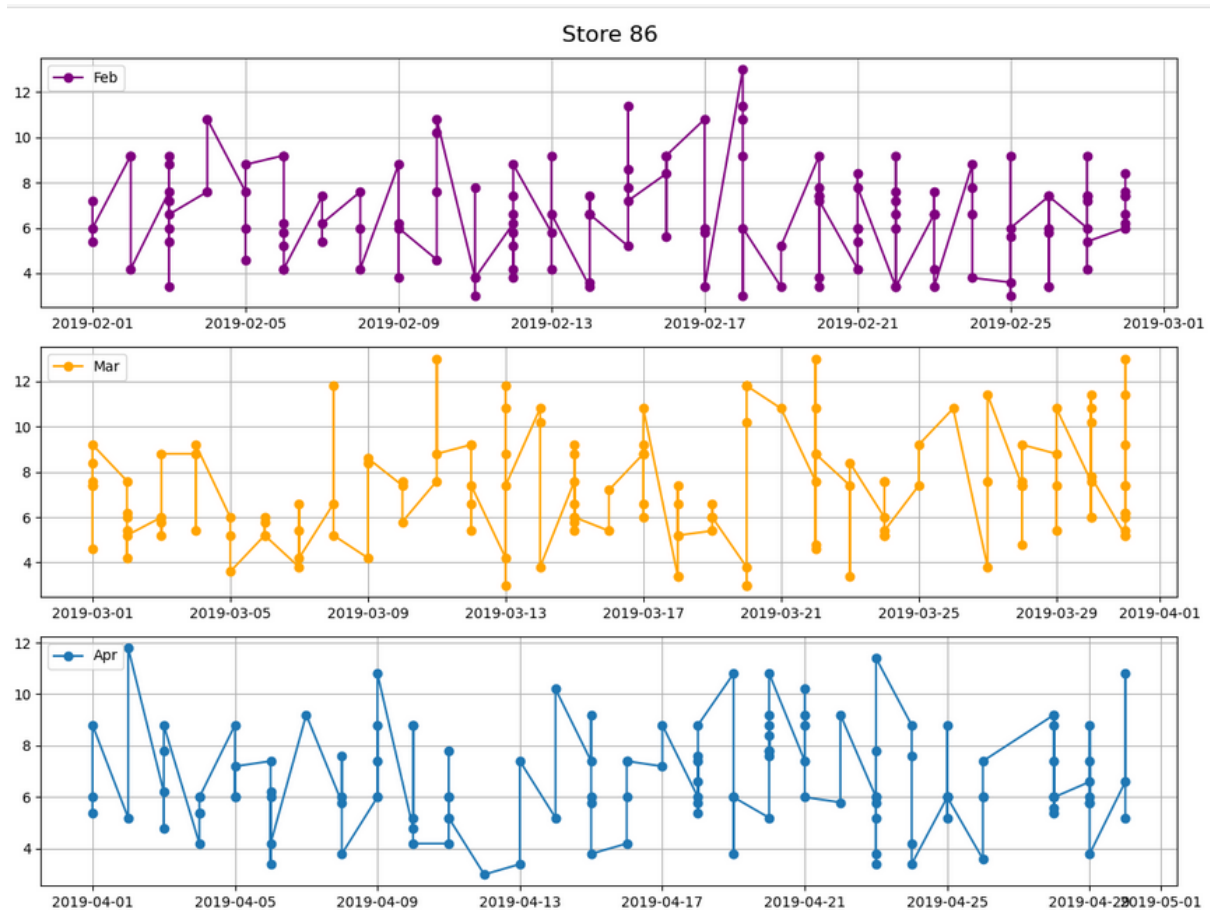
```
ax[2].plot(trail_time_period[(trail_time_period["STORE_NBR"]==86)&(trail_time_period["month"]  
=='Apr')].sort_values(by='DATE')['DATE'],trail_time_period[(trail_time_period["STORE_NBR"]==86  
)&(trail_time_period["month"]=='Apr')].sort_values(by='DATE')['TOT_SALES'], marker='o',  
linestyle='-',label="Apr")
```

```
ax[2].grid(True)
```

```
ax[2].legend(loc='upper left')
```

```
plt.tight_layout()
```

```
plt.show()
```



Store 88

```
fig,ax = plt.subplots(3,1,figsize=(12,9))
```

```
fig.suptitle("Store 88",fontsize=16)
```

```
ax[0].plot(trail_time_period[(trail_time_period["STORE_NBR"]==88)&(trail_time_period["month"]
=='Feb')].sort_values(by='DATE')['DATE'],trail_time_period[(trail_time_period["STORE_NBR"]==8
8)&(trail_time_period["month"]=="Feb")].sort_values(by='DATE')['TOT_SALES'], marker='o',
linestyle='-',color='purple',label='Feb')
```

```
ax[0].grid(True)
```

```
ax[0].legend(loc='upper left')
```

```
ax[1].plot(trail_time_period[(trail_time_period["STORE_NBR"]==88)&(trail_time_period["month"]
=='Mar')].sort_values(by='DATE')['DATE'],trail_time_period[(trail_time_period["STORE_NBR"]==8
```

```
8)&(trail_time_period["month"]=="Mar").sort_values(by='DATE')['TOT_SALES'], marker='o',
linestyle='-',color="orange",label="Mar")
```

```
ax[1].grid(True)
```

```
ax[1].legend(loc='upper left')
```

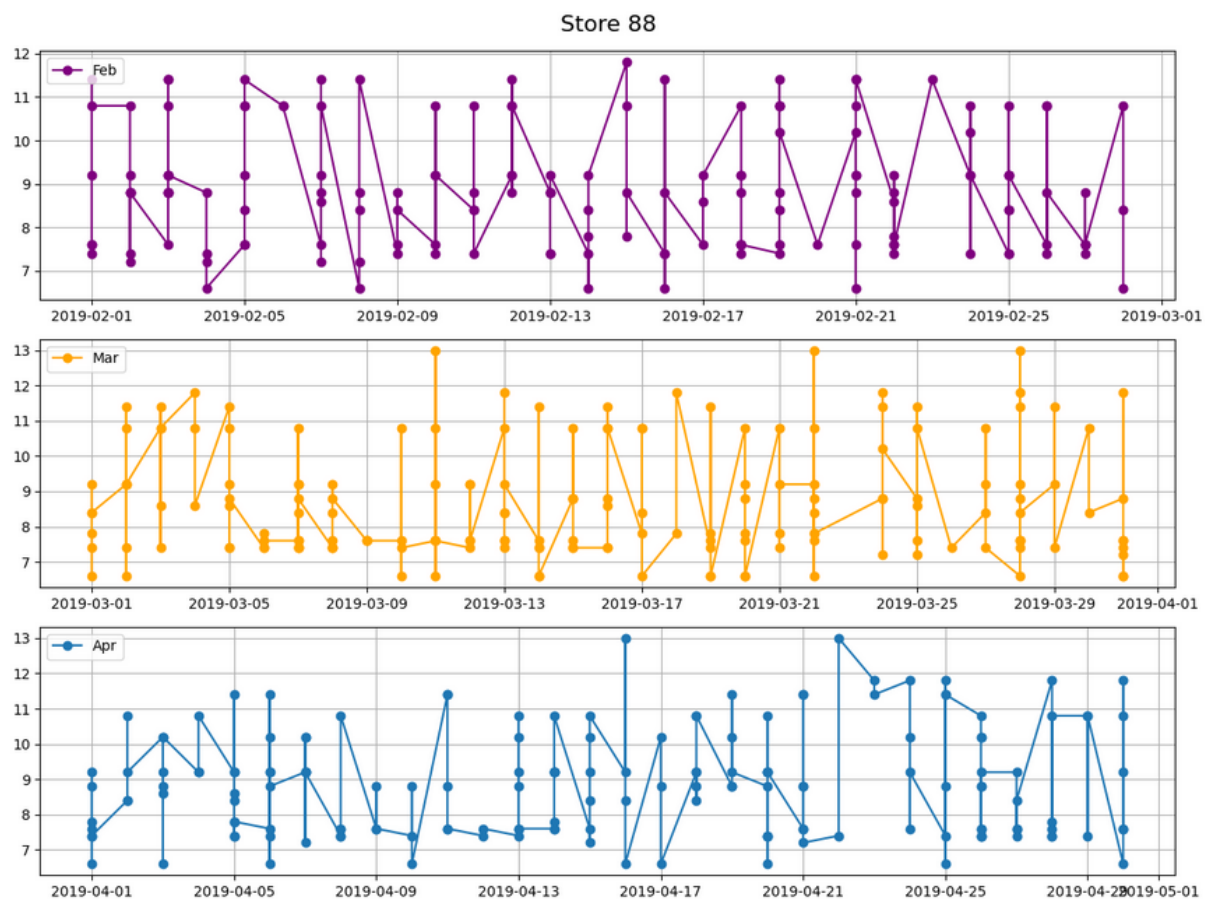
```
ax[2].plot(trail_time_period[(trail_time_period["STORE_NBR"]==88)&(trail_time_period["month"]
=='Apr')].sort_values(by='DATE')['DATE'],trail_time_period[(trail_time_period["STORE_NBR"]==88
)&(trail_time_period["month"]=="Apr")].sort_values(by='DATE')['TOT_SALES'], marker='o',
linestyle='-',label="Apr")
```

```
ax[2].grid(True)
```

```
ax[2].legend(loc='upper left')
```

```
plt.tight_layout()
```

```
plt.show()
```



```
# extracting every trail store 3 months data
```

```
trail_time_period_sorted = trail_time_period.sort_values(by='DATE')
```

```
fig,ax = plt.subplots(3,1,figsize=(12,9))
```

```
fig.suptitle("Store 3 Months sales",fontsize=16)
```

```
ax[0].plot(trail_time_period_sorted[trail_time_period_sorted["STORE_NBR"]==77]['DATE'],trail_time_period_sorted[trail_time_period_sorted["STORE_NBR"]==77]['TOT_SALES'],color='purple',label='store 77')
```

```
ax[0].legend(loc='upper left')
```

```
ax[1].plot(trail_time_period_sorted[trail_time_period_sorted["STORE_NBR"]==86]['DATE'],trail_time_period_sorted[trail_time_period_sorted["STORE_NBR"]==86]['TOT_SALES'],color='orange',label='store 86')
```

```
ax[1].legend(loc='upper left')
```

```
ax[2].plot(trail_time_period_sorted[trail_time_period_sorted["STORE_NBR"]==88]['DATE'],trail_time_period_sorted[trail_time_period_sorted["STORE_NBR"]==88]['TOT_SALES'],label='store 88')
```

```
ax[2].legend(loc='upper left')
```

```
plt.tight_layout()
```

```
plt.show()
```



Revenue visualization store 77,86,88

```
fig,ax = plt.subplots(1,3,figsize=(12,4))
```

```
fig.suptitle("Revnuue",fontsize=16)
```

```
sns.barplot(x=every_month_sales['month'],y=every_month_sales['store_77'],ax=ax[0],label='Store 77')
```

```
ax[0].set_ylabel("")
```

```
ax[0].set_xlabel("")
```

```
ax[0].legend(loc='upper left')
```

```
sns.barplot(x=every_month_sales['month'],y=every_month_sales['store_86'],ax=ax[1],label='Store 86')
```

```
ax[1].set_ylabel("")
```

```
ax[1].set_xlabel("")
```

```
ax[1].legend(loc='upper left')
```

```
sns.barplot(x=every_month_sales['month'],y=every_month_sales['store_88'],ax=ax[2],label='Store 88')
```

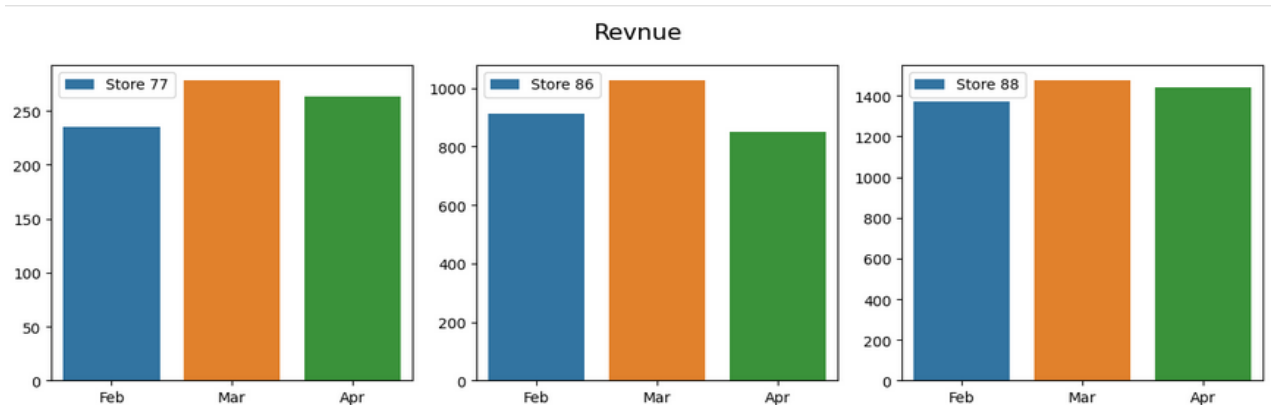
```
ax[2].set_ylabel("")
```

```
ax[2].set_xlabel("")
```

```
ax[2].legend(loc='upper left')
```

```
plt.tight_layout()
```

```
plt.show()
```



```
# Total Revenue by stores
```

```
plt.figure(figsize=(4,3))
```

```
plt.title("Total Reveue by stores",fontsize=10)
```

```
plt.xticks(fontsize=10)
```

```
plt.barh(y=["store_77","store_86","store_88"],  
width=[every_month_sales['store_77'].sum(),every_month_sales['store_86'].sum(),every_month_sales['store_88'].sum()], color=['skyblue', 'lightgreen', 'salmon'])
```

```
# trying to find correlation with trail store data
```

```
all_store = sorted_df[~sorted_df['STORE_NBR'].isin([77,86,88])]
```

```
control_store = all_store[(all_store['DATE']>="2019-02-01")&(all_store['DATE']<="2019-04-30")]
```

```
control_store['Month'] = control_store['DATE'].dt.strftime("%b")
```

```
total_sales = control_store.groupby(['STORE_NBR',  
'Month'])['TOT_SALES'].sum().reset_index(name='total_sales')
```

```
total_customers = control_store.groupby(['STORE_NBR',  
'Month'])['LYLTY_CARD_NBR'].nunique().reset_index(name='total_customers')
```

```
total_transactions = control_store.groupby(['STORE_NBR',  
'Month'])['TXN_ID'].nunique().reset_index(name='total_transactions')
```

```
control_metrics = total_sales.merge(total_customers, on=['STORE_NBR', 'Month'])
```

```
control_metrics = control_metrics.merge(total_transactions, on=['STORE_NBR', 'Month'])
```

```
control_metrics['avg_txn_per_customer'] = control_metrics['total_transactions'] /  
control_metrics['total_customers']
```

```
total_sales_trial = trail_time_period_sorted.groupby(['STORE_NBR',  
'month'])['TOT_SALES'].sum().reset_index(name='total_sales')
```

```
total_customers_trial = trail_time_period_sorted.groupby(['STORE_NBR',  
'month'])['LYLTY_CARD_NBR'].nunique().reset_index(name='total_customers')
```

```
total_transactions_trial = trail_time_period_sorted.groupby(['STORE_NBR',  
'month'])['TXN_ID'].nunique().reset_index(name='total_transactions')
```

```
trial_metrics = total_sales_trial.merge(total_customers_trial, on=['STORE_NBR', 'month'])
```

```
trial_metrics = trial_metrics.merge(total_transactions_trial, on=['STORE_NBR', 'month'])
```

```
trial_metrics['avg_txn_per_customer'] = trial_metrics['total_transactions'] /  
trial_metrics['total_customers']
```

```
# import library to find correlations
```

```
From scipy.stats import pearsonr
```

```
# remaining column name
```

```
control_metrics = control_metrics.rename(columns={'Month': 'month'})
```

```

def compare_stores_metric(trial_df, control_df, metric='total_sales'):

    # Merge on month to align months

    merged = trial_df.merge(control_df, on='month', suffixes=('_trial', '_control'))

    if len(merged) < 2:

        print("Not enough overlapping months between trial and control store.")

        return None, None

    try:

        corr, _ = pearsonr(merged[f'{metric}_trial'], merged[f'{metric}_control'])

        obs_distance = np.linalg.norm(

            merged[f'{metric}_trial'].values - merged[f'{metric}_control'].values

        )

        return corr, obs_distance

    except Exception as e:

        print(f"Error comparing stores: {e}")

        return None, None

#trying to find correlation for store 77

trial_store_nbr = 77

trial_store_data = trial_metrics[trial_metrics['STORE_NBR'] == trial_store_nbr]

control_store_nbrs = control_metrics['STORE_NBR'].unique()

results = []

for ctrl_store in control_store_nbrs:

    control_store_data = control_metrics[control_metrics['STORE_NBR'] == ctrl_store]

```

```
corr, dist = compare_stores_metric(trial_store_data, control_store_data, metric='total_sales')
```

```
if corr is not None:
```

```
    results.append({  
        'trial_store': trial_store_nbr,  
        'control_store': ctrl_store,  
        'pearson_corr': corr,  
        'magnitude_distance': dist  
    })
```

```
distances = [r['magnitude_distance'] for r in results]
```

```
min_dist = min(distances)
```

```
max_dist = max(distances)
```

```
for r in results:
```

```
    if max_dist != min_dist:
```

```
        r['magnitude_similarity'] = 1 - ((r['magnitude_distance'] - min_dist) / (max_dist - min_dist))
```

```
    else:
```

```
        r['magnitude_similarity'] = 1.0
```

```
# finding max pearson_corr
```

```
max(results, key=lambda x: x['pearson_corr'])
```

```
output -> {'trial_store': 77,
```

```
         'control_store': 26,
```

```
         'pearson_corr': 0.9999036297409183,
```

```
         'magnitude_distance': 1655.8964762327385,
```

```
         'magnitude_similarity': 0.19341259440383818}
```

```
# Then taking talking trail store 77 and control store 26 and conducting T-test for total sales
```

```
t_stat, p_value =  
ttest_ind(control_metrics[control_metrics['STORE_NBR']==26]['total_sales'],trial_metrics[trial_  
metrics['STORE_NBR']==77]['total_sales'])
```

```
#trying to find correlation for store 86
```

```
trial_store_nbr = 86
```

```
trial_store_data = trial_metrics[trial_metrics['STORE_NBR'] == trial_store_nbr]
```

```
results = []
```

```
for ctrl_store in control_store_nbrs:
```

```
    control_store_data = control_metrics[control_metrics['STORE_NBR'] == ctrl_store]
```

```
    corr, dist = compare_stores_metric(trial_store_data, control_store_data, metric='total_sales')
```

```
    if corr is not None:
```

```
        results.append({  
            'trial_store': trial_store_nbr,  
            'control_store': ctrl_store,  
            'pearson_corr': corr,  
            'magnitude_distance': dist  
        })
```

```
distances = [r['magnitude_distance'] for r in results]
```

```
min_dist = min(distances)
```

```
max_dist = max(distances)
```

```
for r in results:
```

```
    if max_dist != min_dist:
```

```
        r['magnitude_similarity'] = 1 - ((r['magnitude_distance'] - min_dist) / (max_dist - min_dist))
```

```
    else:
```

```
        r['magnitude_similarity'] = 1.0
```

```
max(results, key=lambda x: x['pearson_corr'])
```

```
output -> {'trial_store': 86,
```

```
         'control_store': 171,
```

```
         'pearson_corr': 0.9998543299829847,
```

```
         'magnitude_distance': 908.3676403307198,
```

```
         'magnitude_similarity': 0.4511653145431007}
```

```
# Then taking talking trail store 86 and control store 171 and conducting T-test for total sales
```

```
t_stat, p_value =
```

```
ttest_ind(control_metrics[control_metrics['STORE_NBR']==171]['total_sales'], trial_metrics[trial_metrics['STORE_NBR']==86]['total_sales'])
```

```
#trying to find correlation for store 88
```

```
trial_store_nbr = 88
```

```
trial_store_data = trial_metrics[trial_metrics['STORE_NBR'] == trial_store_nbr]
```

```
results = []
```

```
for ctrl_store in control_store_nbrs:
```

```
    control_store_data = control_metrics[control_metrics['STORE_NBR'] == ctrl_store]
```

```
corr, dist = compare_stores_metric(trial_store_data, control_store_data, metric='total_sales')
```

```
if corr is not None:
```

```
    results.append({  
        'trial_store': trial_store_nbr,  
        'control_store': ctrl_store,  
        'pearson_corr': corr,  
        'magnitude_distance': dist  
    })
```

```
distances = [r['magnitude_distance'] for r in results]
```

```
min_dist = min(distances)
```

```
max_dist = max(distances)
```

```
for r in results:
```

```
    if max_dist != min_dist:
```

```
        r['magnitude_similarity'] = 1 - ((r['magnitude_distance'] - min_dist) / (max_dist - min_dist))
```

```
    else:
```

```
        r['magnitude_similarity'] = 1.0
```

```
max(results, key=lambda x: x['pearson_corr'])
```

```
output -> {'trial_store': 88,
```

```
         'control_store': 26,
```

```
         'pearson_corr': 0.9999902269288453,
```

```
         'magnitude_distance': 384.6506986864836,
```

```
'magnitude_similarity': 0.9000892094661825}
```

```
# Then taking talking trail store 88 and control store 26 and conducting T-test for total sales
```

```
t_stat, p_value =  
ttest_ind(control_metrics[control_metrics['STORE_NBR']==26]['total_sales'],trial_metrics[trial_  
metrics['STORE_NBR']==88]['total_sales'])
```

```
# Then taking talking trail store 77 and control store 26 and conducting T-test for total customer
```

```
t_stat, p_value =  
ttest_ind(control_metrics[control_metrics['STORE_NBR']==26]['total_customers'],trial_metrics[  
trial_metrics['STORE_NBR']==77]['total_customers'])
```

```
# Then taking talking trail store 86 and control store 171 and conducting T-test for total  
customer
```

```
t_stat, p_value =  
ttest_ind(control_metrics[control_metrics['STORE_NBR']==171]['total_customers'],trial_metrics  
[trial_metrics['STORE_NBR']==86]['total_customers'])
```

```
# Then taking talking trail store 88 and control store 26 and conducting T-test for total customer
```

```
t_stat, p_value =  
ttest_ind(control_metrics[control_metrics['STORE_NBR']==26]['total_customers'],trial_metrics[  
trial_metrics['STORE_NBR']==88]['total_customers'])
```

```
# Then taking talking trail store 77 and control store 26 and conducting T-test for  
avg_txn_per_customer
```

```
t_stat, p_value =  
ttest_ind(control_metrics[control_metrics['STORE_NBR']==26]['avg_txn_per_customer'],trial_m  
etrics[trial_metrics['STORE_NBR']==77]['avg_txn_per_customer'])
```

```
# Then taking talking trail store 86 and control store 171 and conducting T-test for  
avg_txn_per_customer
```

```
t_stat, p_value =  
ttest_ind(control_metrics[control_metrics['STORE_NBR']==171]['avg_txn_per_customer'],trial_  
metrics[trial_metrics['STORE_NBR']==86]['avg_txn_per_customer'])
```

```
# Then taking talking trail store 88 and control store 26 and conducting T-test for  
avg_txn_per_customer
```

```
t_stat, p_value =  
ttest_ind(control_metrics[control_metrics['STORE_NBR']==26]['avg_txn_per_customer'],trial_m  
etrics[trial_metrics['STORE_NBR']==88]['avg_txn_per_customer'])
```

```
pearson relation data
```

```
data = {
```

```
"store":["storeT77 - storeC26","storeT86 - storeC171","storeT88 - storeC26"],
```

```
'total_customer -p-value':[0.0009091423244386907,7.066041850199187e-  
05,0.24480233660157946],
```

```
'total_sales -p-  
value':[0.04029325439448244,0.0008788706534906513,0.04029325439448244],
```

```
'avg_txn_per_customer -p-  
value':[0.030165522872758223,0.0028061257864853482,0.020518617844872907],
```

```
'pearson_corr': [0.9999036297409183,0.9998543299829847,0.9999902269288453],
```

```
'magnitude_distance': [1655.8964762327385,908.3676403307198,384.6506986864836],
```

```
'magnitude_similarity': [0.19341259440383818,0.4511653145431007,0.9000892094661825]
```

```
}
```

Checking behaviour changed

```
if data['total_customer -p-value'][0]<0.05:
```

```
    print("Behaviour has changed")
```

```
else:
```

```
    print("Behaviour has not changed")
```

```
if data['total_customer -p-value'][1]<0.05:
```

```
    print("Behaviour has changed")
```

```
else:
```

```
    print("Behaviour has not changed")
```

```
if data['total_customer -p-value'][2]<0.05:
```

```
    print("Behaviour has changed")
```

```
else:
```

```
    print("Behaviour has not changed")
```

```
if data['total_sales -p-value'][0]<0.05:
```

```
    print("Behaviour has changed")
```

```
else:
```

```
    print("Behaviour has not changed")
```

```
if data['total_sales -p-value'][1]<0.05:
```

```
    print("Behaviour has changed")
```

```
else:
```

```
    print("Behaviour has not changed")
```

```
if data['total_sales -p-value'][2]<0.05:
```

```
    print("Behaviour has changed")
```

```
else:
```

```
    print("Behaviour has not changed")
```

```
if data['avg_txn_per_customer -p-value'][0]<0.05:
```

```
    print("Behaviour has changed")
```

```
else:
```

```
    print("Behaviour has not changed")
```

```
if data['avg_txn_per_customer -p-value'][1]<0.05:
```

```
    print("Behaviour has changed")
```

```
else:
```

```
    print("Behaviour has not changed")
```

```
if data['avg_txn_per_customer -p-value'][2]<0.05:
```

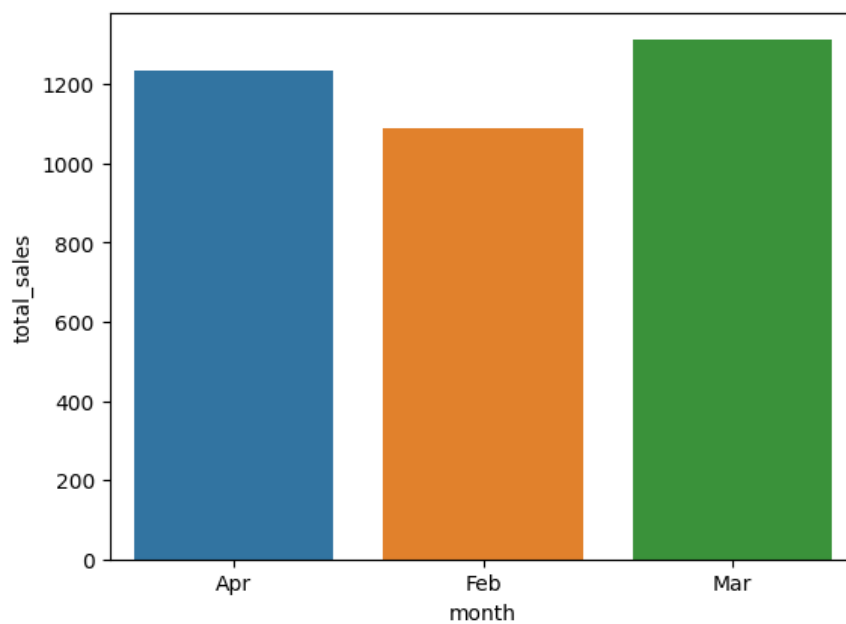
```
    print("Behaviour has changed")
```

```
else:
```

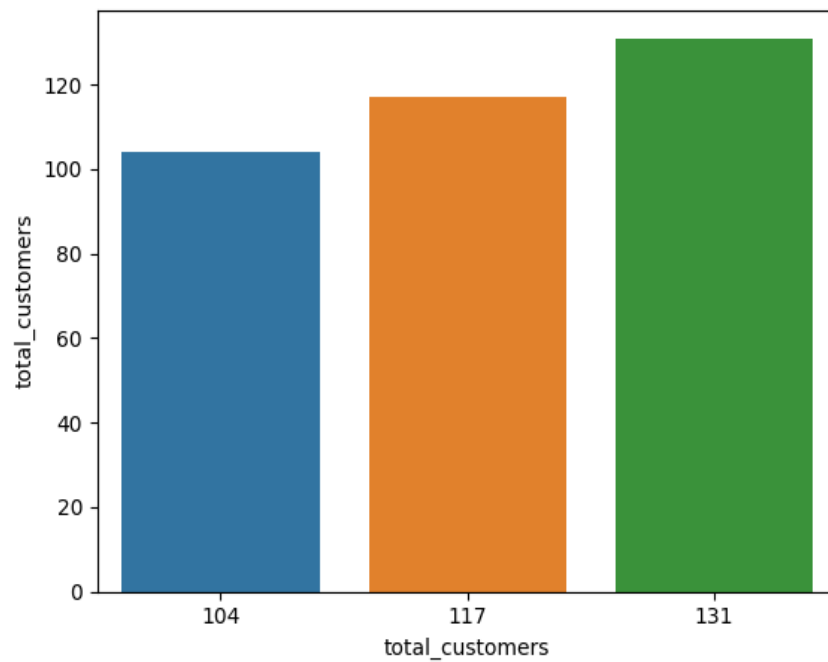
```
    print("Behaviour has not changed")
```

```
# plotting control store 26 total sales , total customer , average transaction
```

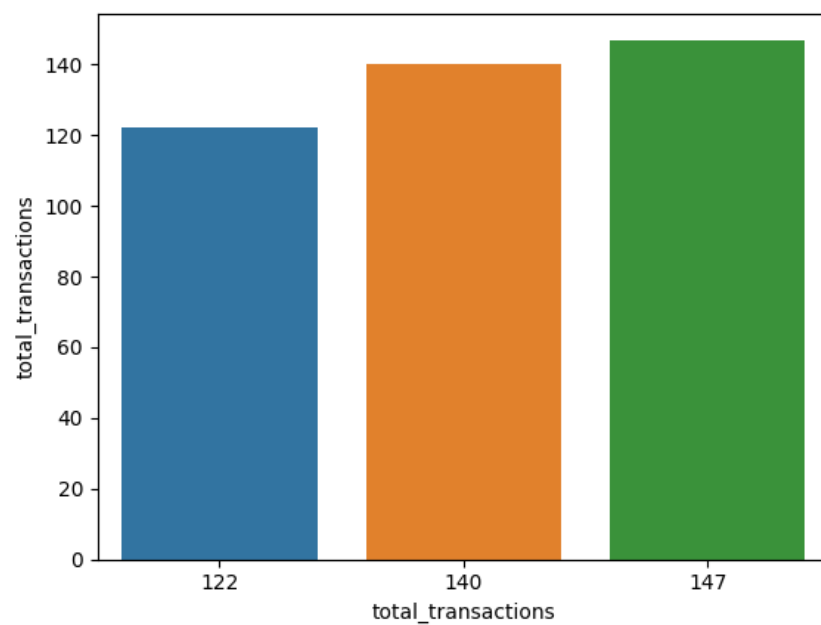
```
sns.barplot(x=control_metrics[control_metrics['STORE_NBR']==26]['month'],y=control_metrics[control_metrics['STORE_NBR']==26]['total_sales'])
```



```
sns.barplot(x=control_metrics[control_metrics['STORE_NBR']==26]['month'],y=control_metrics[control_metrics['STORE_NBR']==26]['total_sales'])
```

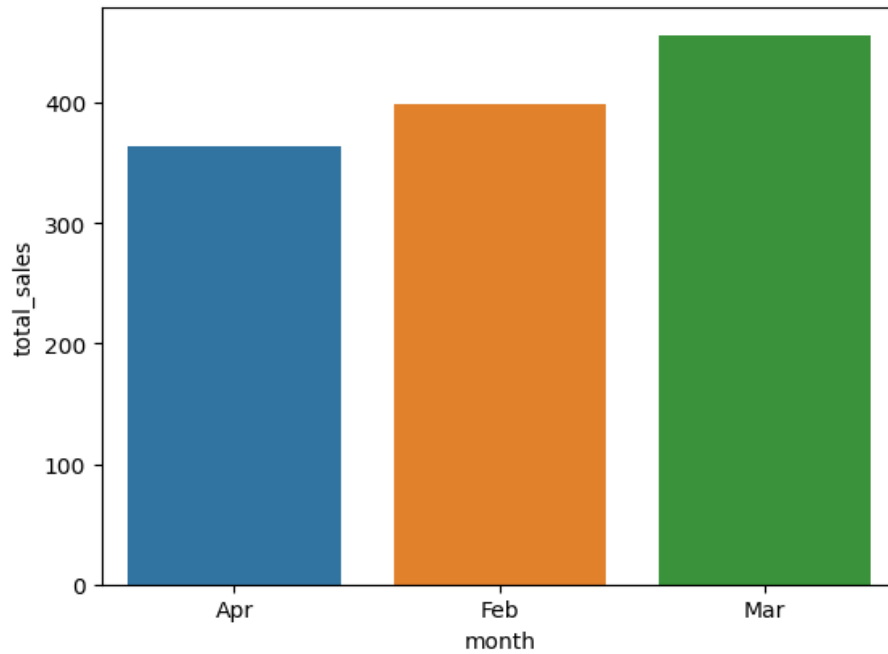


```
sns.barplot(x=control_metrics[control_metrics['STORE_NBR']==26]['total_customers'],y=control_metrics[control_metrics['STORE_NBR']==26]['total_customers'])
```

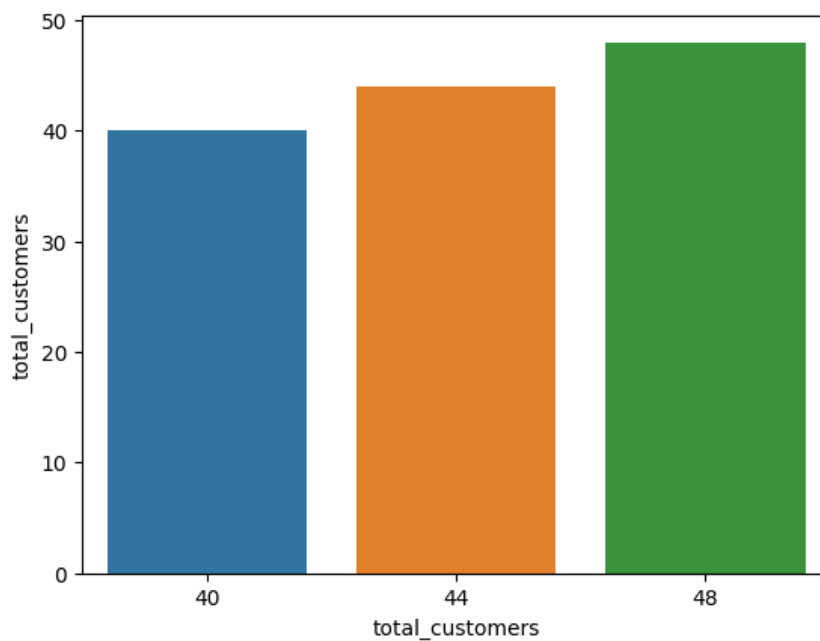


```
# plotting control store 171 total sales , total customer , average transaction
```

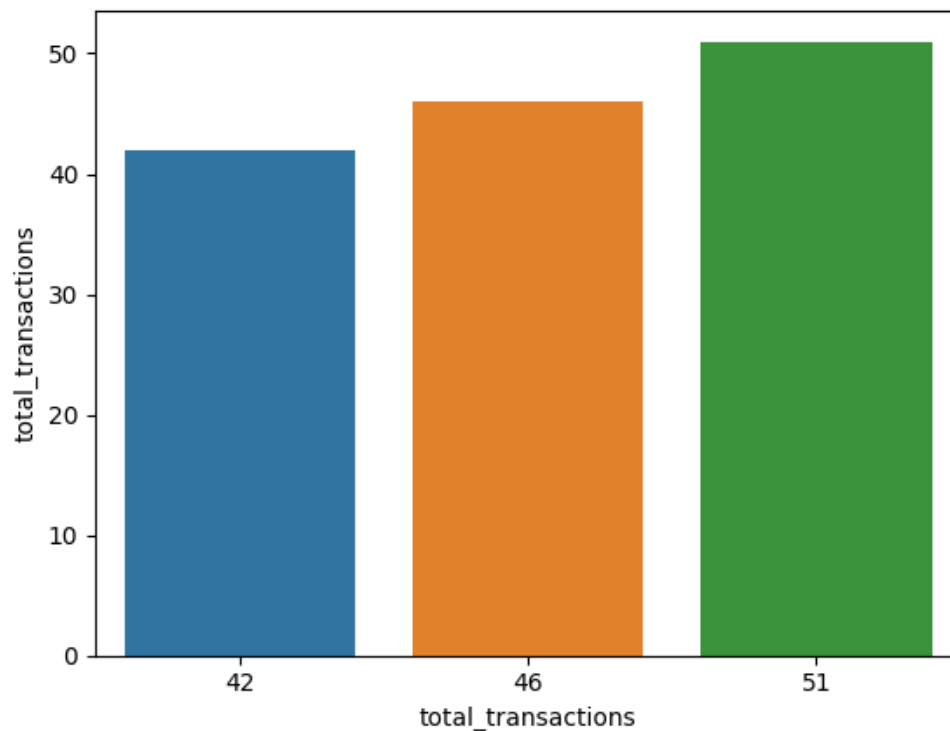
```
sns.barplot(x=control_metrics[control_metrics['STORE_NBR']==171]['month'],y=control_metrics[control_metrics['STORE_NBR']==171]['total_sales'])
```



```
sns.barplot(x=control_metrics[control_metrics['STORE_NBR']==171]['total_customers'],y=control_metrics[control_metrics['STORE_NBR']==171]['total_customers'])
```



```
sns.barplot(x=control_metrics[control_metrics['STORE_NBR']==171]['total_transactions'],y=control_metrics[control_metrics['STORE_NBR']==171]['total_transactions'])
```



```
# Plotting store control store 26 and store 171 trend sales
```

```
# every month wise
```

```
fig,ax = plt.subplots(3,1,figsize=(12,9))
```

```
fig.suptitle("Store 26",fontsize=16)
```

```
ax[0].plot(store_26[store_26['month']=='Feb']['DATE'],store_26[store_26['month']=='Feb']['TOT_SALES'],label='Feb',color='purple',marker='o', linestyle='-')
```

```
ax[0].grid(True)
```

```
ax[0].legend(loc='upper left')
```

```
ax[1].plot(store_26[store_26['month']=='Mar']['DATE'],store_26[store_26['month']=='Mar']['TOT_SALES'], marker='o', linestyle='-',color="orange",label="Mar")
```

```
ax[1].grid(True)
```

```
ax[1].legend(loc='upper left')
```

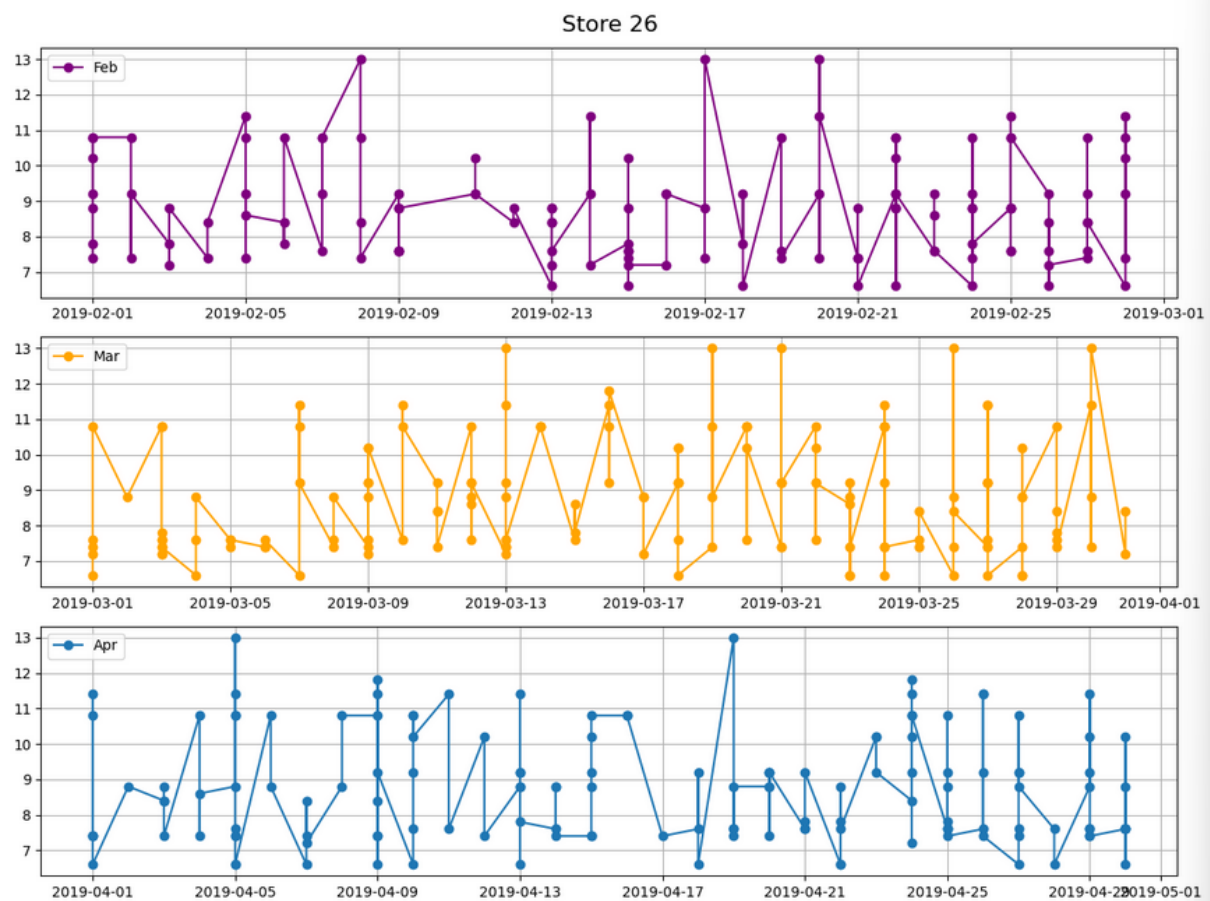
```
ax[2].plot(store_26[store_26['month']=='Apr']['DATE'],store_26[store_26['month']=='Apr']['TOT_SALES'], marker='o', linestyle='-',label="Apr")
```

```
ax[2].grid(True)
```

```
ax[2].legend(loc='upper left')
```

```
plt.tight_layout()
```

```
plt.show()
```



```
fig,ax = plt.subplots(3,1,figsize=(12,9))
```

```
fig.suptitle("Store 171",fontsize=16)
```

```
ax[0].plot(store_171[store_171['month']=='Feb']['DATE'],store_171[store_171['month']=='Feb']['TOT_SALES'],label='Feb',color='purple',marker='o', linestyle='-')
```

```
ax[0].grid(True)
```

```
ax[0].legend(loc='upper left')
```

```
ax[1].plot(store_171[store_171['month']=='Mar']['DATE'],store_171[store_171['month']=='Mar']['TOT_SALES'], marker='o', linestyle='-',color="orange",label="Mar")
```

```
ax[1].grid(True)
```

```
ax[1].legend(loc='upper left')
```

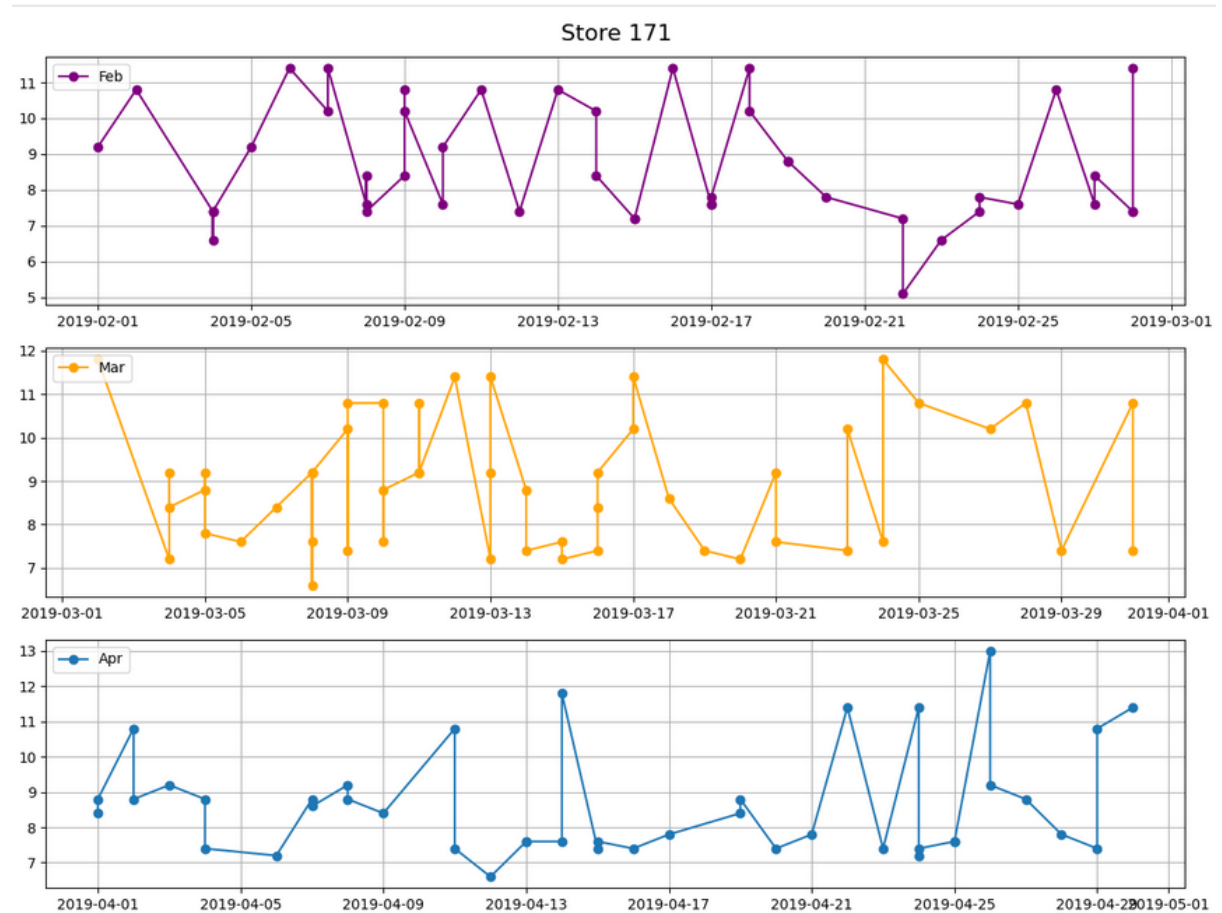
```
ax[2].plot(store_171[store_171['month']=='Apr']['DATE'],store_171[store_171['month']=='Apr']['TOT_SALES'], marker='o', linestyle='-',label="Apr")
```

```
ax[2].grid(True)
```

```
ax[2].legend(loc='upper left')
```

```
plt.tight_layout()
```

```
plt.show()
```



```
#combine 3 months
```

```
fig,ax = plt.subplots(2,1,figsize=(12,7))
```

```
fig.suptitle("Store 3 Months sales",fontsize=16)
```

```
ax[0].plot(store_26['DATE'],store_26['TOT_SALES'],color='purple',label='store 26')
```

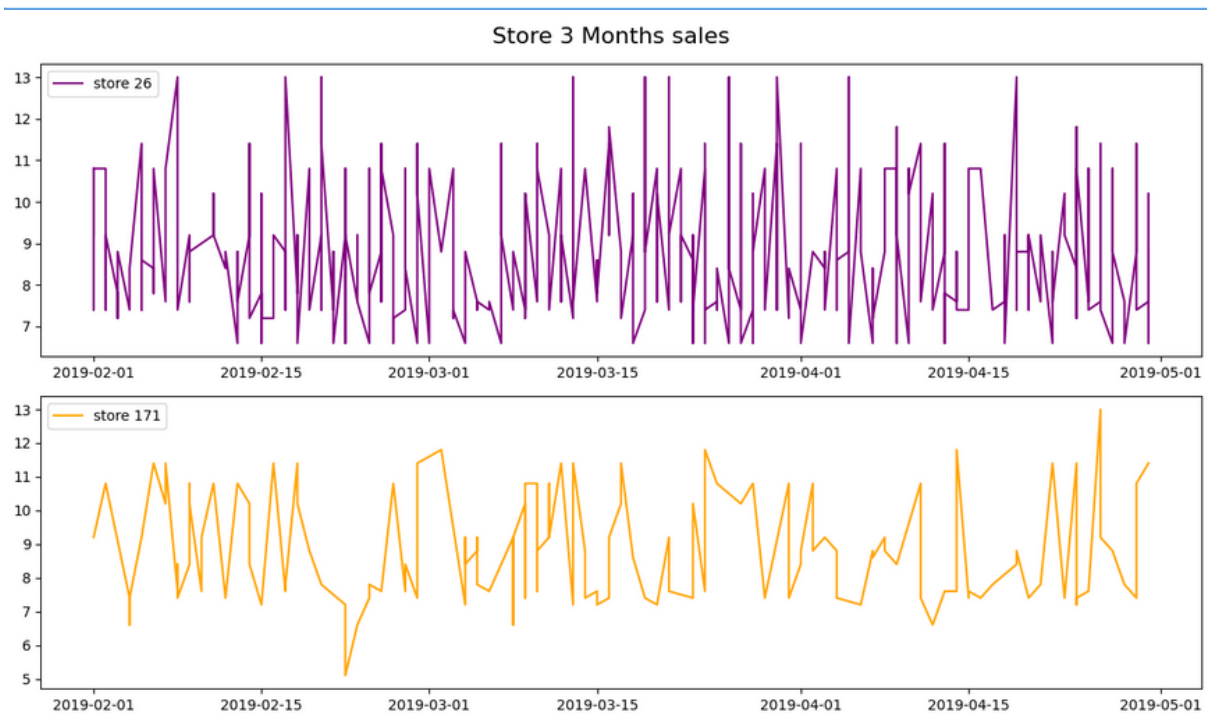
```
ax[0].legend(loc='upper left')
```

```
ax[1].plot(store_171['DATE'],store_171['TOT_SALES'],color='orange',label='store 171')
```

```
ax[1].legend(loc='upper left')
```

```
plt.tight_layout()
```

```
plt.show()
```



```
# Total Revenue by stores
```

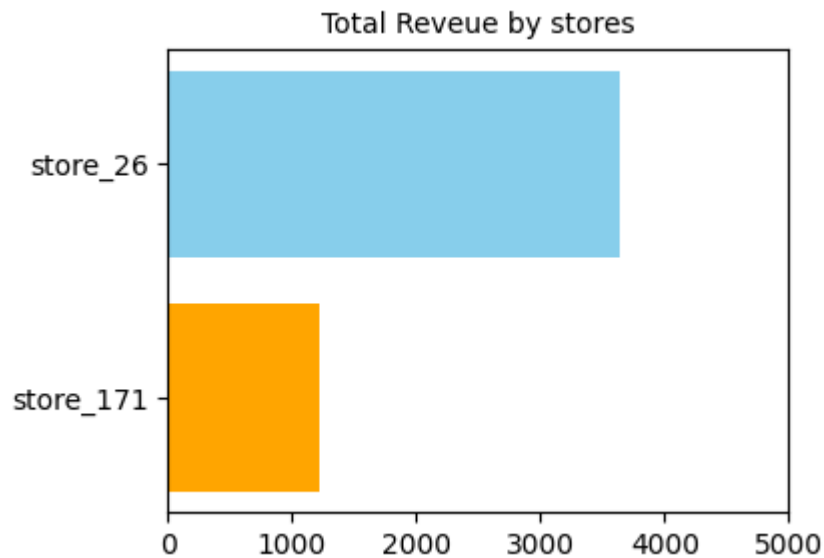
```
plt.figure(figsize=(4,3))
```

```
plt.title("Total Reveue by stores",fontsize=10)
```

```
plt.xticks(fontsize=10)
```

```
plt.xlim(0,5000)
```

```
plt.barh(["store_171","store_26"],[store_171['TOT_SALES'].sum(),store_26['TOT_SALES'].sum()],color=['orange','skyblue'])
```



Conclusion: -

- Total sales, total customer, average customer transaction behaviour has changed.
- Store 26 ,171 control store is best co relation with our trail data according to peasonr relation
- In store 86 in March month, they have generated better Revenue compared to other months.
- Ploting all trail store revenue for visualization perspective.
- Generating two separate set trail store and control store
- Using t-test method to find relationships
- Storing all p-value and other matrix storing in dictionary format and storing in data variables
- Ploting control store 26,171 sales trend, total customer etc.